

Instructions:

1. Attempt both sections in separate answer books.
2. Figure to right indicates marks.
3. Make suitable assumption when necessary

Section I

Q1. (A) Answer Following. [05]

1. _____ level of database describes how the record is stored.
2. Managing locks on data items is responsibility of_____.
3. List down four applications of database management system.
4. List down any 4 RDBMS applications.
5. Algorithm for Recovery and Isolation Exploiting Semantics is used for_____.

Q1. (B) Answer Following. [06]

1. List and explain different levels of DBMS abstraction.
2. List and explain different storage structures for DBMS.
3. Give full form of following: 1) DDL; 2) DML.

Q2. Answer Following. [12]

1. Explain check-pointing technique of transaction recovery.
2. Explain atomicity and consistency properties.

OR

Q2. Answer Following. [12]

1. Explain states of transaction.
2. Explain isolation and durability properties.

Q3. Answer Following. [12]

1. What is transaction? Explain transaction concept with example.
2. Explain different components of distributed database transaction.

OR

Q3. Answer Following. [12]

1. What is database recovery? Explain different types of recovery techniques.
2. Write a short note on two phase commit protocol.

SECTION II

Q4. (A) Do as Directed [05]

1. ____ deadlock detection technique faces overhead problem.
2. In ____ protocol in the absence of any information about a transaction, the recovery process orders an inquiring subordinate to abort, assuming that the transaction has been aborted.
3. Give true/false: R * provides location transparency.
4. Which data structure is used by system to detect deadlock?
5. Each distributed database system has ____, which is used for recovery.

Q4. (B) Answer Following. [06]

1. Why do we need parallel databases?
2. Give full form of GIS. List down components of GIS.
3. What is an active database? What are its different applications?

Q5. Do as directed [12]

1. What are advantages and disadvantages of parallel database?
2. Write a short note on shared memory system.

OR

Q5. Answer Following. [12]

1. What is partition parallelism? Explain round robin technique of query evaluation.
2. Write a short note on shared nothing system.

Q6. Answer Following [12]

1. What is query evaluation in parallel databases? Explain pipeline parallel query evaluation technique.
2. Write a short note on shared disk system.

OR

Q6. Answer Following. [12]

1. Explain following parallel query optimization techniques: 1) Hash Partition; 2) Range Partition.
2. Write a short note on parallel query optimization.